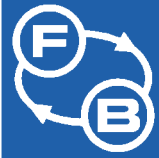


Control in a MATLAB Environment
Introduction and Computer Systems Installation
33-000V61



Feedback



E-Learning



Electricity & Electronics



Control & Instrumentation



Process Control



Mechatronics



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Technology Training for tomorrow's world

Control in a MATLAB[®] Environment

Introduction and Computer Systems Installation

33-000V61

**(For use with MATLAB 6.1 version
and Windows 95, 98, NT, 2000)**



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Notes



INTRODUCTION AND SOFTWARE INSTALLATION

Preface

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We are required under the Health and Safety at Work Act 1974, to make available to users of this equipment certain information regarding its safe use.

The equipment, when used in normal or prescribed applications within the parameters set for its mechanical and electrical performance, should not cause any danger or hazard to health or safety if normal engineering practices are observed and they are used in accordance with the instructions supplied.

If, in specific cases, circumstances exist in which a potential hazard may be brought about by careless or improper use, these will be pointed out and the necessary precautions emphasised.

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This equipment should not be used by inexperienced users unless they are under supervision.

We are required by European Directives to indicate on our equipment panels certain areas and warnings that require attention by the user. These have been indicated in the specified way by yellow labels with black printing, the meaning of any labels that may be fixed to the instrument are shown below:



CAUTION -
RISK OF
DANGER



CAUTION -
RISK OF
ELECTRIC SHOCK



CAUTION -
ELECTROSTATIC
SENSITIVE DEVICE

Refer to accompanying documents

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We maintain a policy of continuous product improvement by incorporating the latest developments and components into our equipment, even up to the time of dispatch.

All major changes are incorporated into up-dated editions of our manuals and this manual was believed to be correct at the time of printing. However, some product changes which do not affect the instructional capability of the equipment, may not be included until it is necessary to incorporate other significant changes.

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- | | |
|------------------------|----------------------------|
| 1. Equipment type | 2. Component value |
| 3. Component reference | 4. Equipment serial number |

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INTRODUCTION AND COMPUTER SYSTEMS INSTALLATION

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Should this equipment be used outside the classroom, laboratory study area or similar such place for which it is designed and sold then Feedback Instruments Ltd hereby states that conformity with the protection requirements of the European Community Electromagnetic Compatibility Directive (89/336/EEC) may be invalidated and could lead to prosecution.

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INTRODUCTION AND COMPUTER SYSTEMS INSTALLATION

Contents

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1 Introduction

The material presented in these manuals is designed to demonstrate the operation of digital control in number of control situations. A complete system consists of a PC, MATLAB and associated toolboxes and utilities and one of a number of different hardware devices designed to show how control operates in real live situation physical situations.

The curriculum discusses the desired control objective, the device used to demonstrate it and the control methodology to be employed.

Examples are given showing how the controller is implemented graphically using MATLAB and Simulink.

Practical work is included which demonstrates the operation of the controller on the device and allows the student to change certain of the control parameters to allow for differences in hardware characteristics and improve the functioning of the controller.

The Environment

The controller is implemented on a PC using software tools from Mathworks Inc. These include:

MATLAB

Simulink

Real Time Workshop

RTW

Real Time Windows Target

RTWT

Visual C++ Professional

(from Microsoft)

all operating in a **Windows** environment.



MATLAB

MATLAB acts as the application Host environment in which the other Mathworks products run. It provides a sophisticated set of tools for solving mathematical problems. In addition there are specialised tool boxes which extend the Matlab functions in several different specific application areas.

Simulink

Simulink is a graphics based system for modelling processes (in our case **Control**), which takes the form of Blocks linked by connecting lines which show how output data from one block is fed as input into another block. Blocks perform specialised operations on the data and may be standard blocks from the Simulink Library or written by the user where no suitable library block exists. The Simulink model is then passed to Real Time Workshop.

Real Time Workshop - RTW

Real Time Workshop automatically builds a C++ source program from the Simulink model.

C++ Compiler

The C++ compiler compiles and links the code created by Real Time Workshop to produce an executable program. The program interfaces to the outside environment via a "Target", in our case Real Time Windows Target.

Real Time Windows Target

Real Time Windows Target communicates with the executable program acting as the Control program, and interfaces with the hardware device through an I/O board.



Real Time Windows Target controls the two way data or signal flow to and from the model (now an executable program), and to and from the I/O Board.

When the program is running, the user may change certain of the parameters in the Simulink model which are then passed, via Real Time Workshop, to the executable program. Any change in the structure of the model, however, will require Real Time Workshop to re-build the program.

Relationship between the elements of the environment

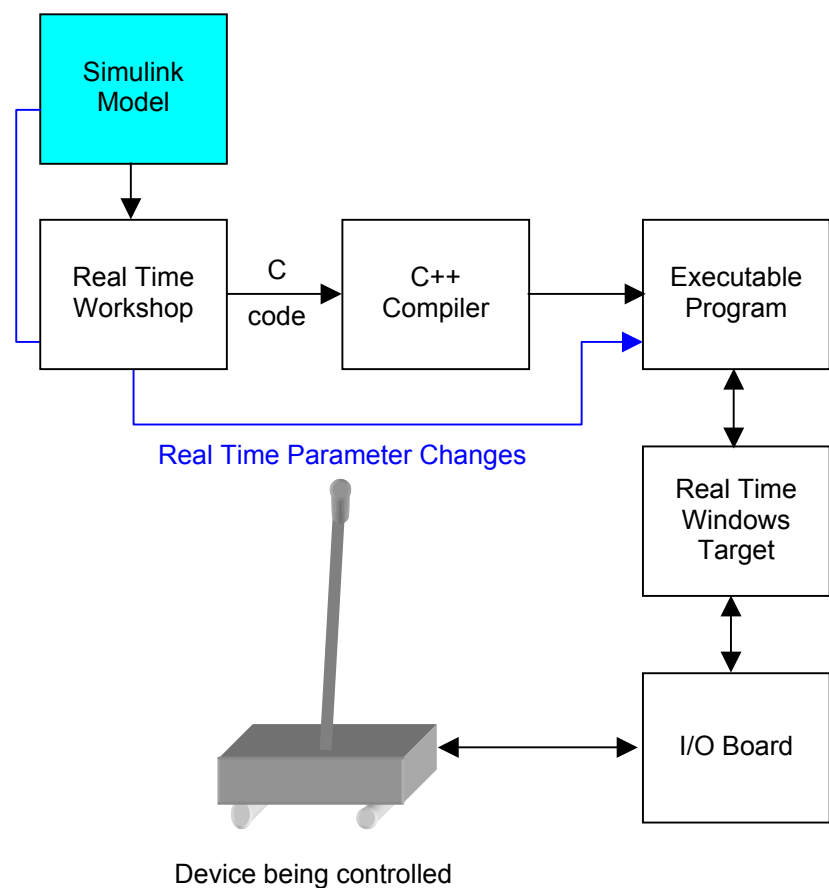


Figure 1-1: Control System Development Flow diagram



Figure 1-1 shows how the various elements of the MATLAB environment link together to provide an integrated set of tools for control system design and experimental validation.

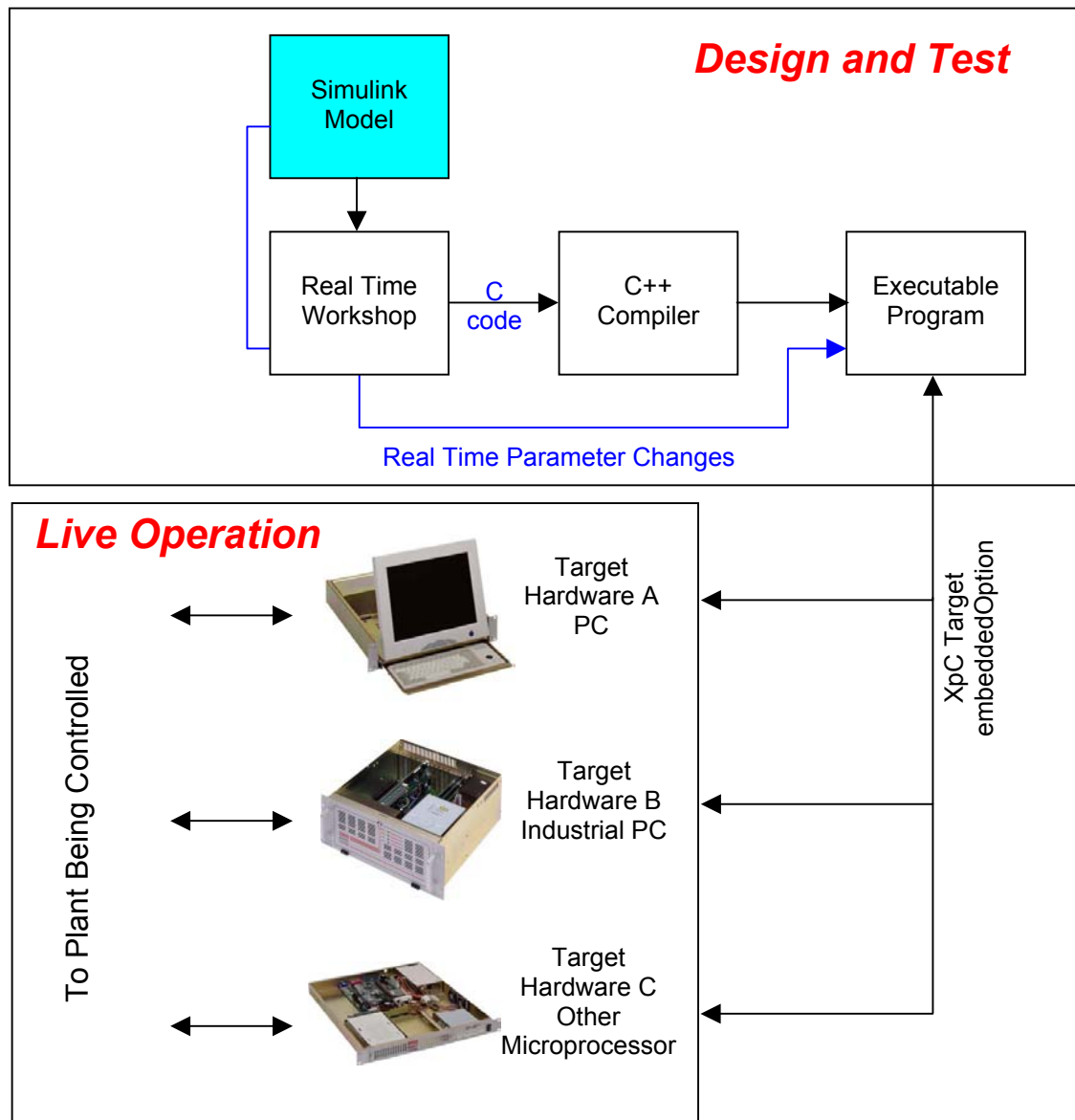


Figure 1-2



Advantages of this approach

The use of standard software tools means that, during the controller design stage, the designer only needs to model the process using the graphics tools available in Simulink (the shaded box in Figure 1-1), without being concerned with the mechanics of communication to and from the device under test.

The tools described above, produced by the Mathworks, are in very widespread use across a number of Industries for control system development. The control engineering student is very likely to meet them later during his or her career, so that familiarity with their use is a valuable addition to the student's skillset.

Figure 1-2 shows use of the tools to create an industrial controller. An additional product from Mathworks Inc, xPC, provides a mechanism for running the executable control code on a PC without Windows, thus avoiding unnecessary software complexity and overhead.

These PC's can be either standard or industrial PC's depending on the application, environment and degree of redundancy required. In addition code can be generated for other hardware architectures than PC.



Feedback Control Products

Feedback Instruments manufacture a number of products for use within the MATLAB environment.

They are the following electromechanical plants:

Digital Pendulum

The Digital Pendulum is used to demonstrate control of a SIMO (Single Input Multiple Output) system. Consisting of a cart moving on an elevated track, and with two free swinging arms attached, it is used to demonstrate:

an inverted pendulum, which illustrates control of an unstable system. Unstable systems are found in controlling rockets at take off, and in humanoid robotics research.

a suspended pendulum which illustrates the sort of control a gantry crane might require when moving, for example, a container of molten steel with minimum oscillation of the load.



Figure 1-3: Digital Pendulum System



Twin Rotor MIMO System (Helicopter)



Figure 1-4: Twin Rotor MIMO System

The Twin Rotor Helicopter is used to demonstrate control of a MIMO (Multiple Input Multiple Output) system. It consists of a vertical and a horizontal rotor fixed to an arm which can rotate about both a vertical axis and a horizontal axis.

As in a real helicopter, two rotors are required for stable operation to counter the reaction on the helicopter body of changes in the angular momentum of the main rotor.

Magnetic Levitation System (Maglev)



Figure 1-5: Magnetic Levitation System



The Magnetic Levitation system demonstrates control of a metal sphere suspended in a magnetic field counteracting the force of gravity. It is an example of a system which is linear between only a small difference between the lower and upper limits of the sphere's vertical travel. Outside of these limits the system becomes highly non linear and impossible to control.

Modular Servo System

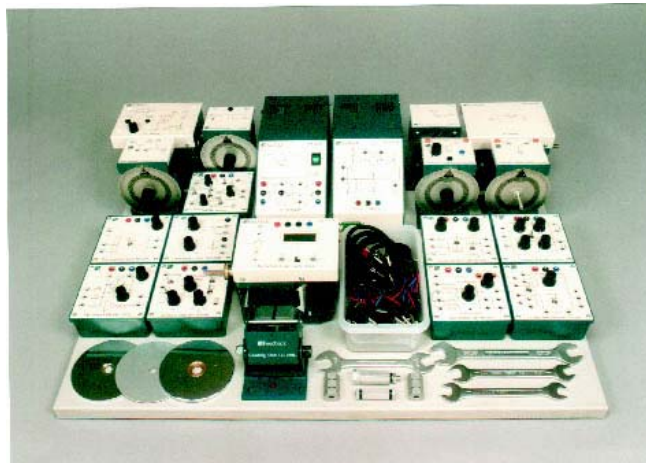


Figure 1-6: Modular Servo System

The Modular Servo system is a set of precision units which may be configured into a fully featured servo system.

Comprising power supply, dc motor, amplifiers, input and output potentiometers, digital encoder, tachometer, motor load unit and PC interface unit, the Modular Servo System provides an open architecture system for accurately demonstrating motor position and speed control.



Digital Servo System

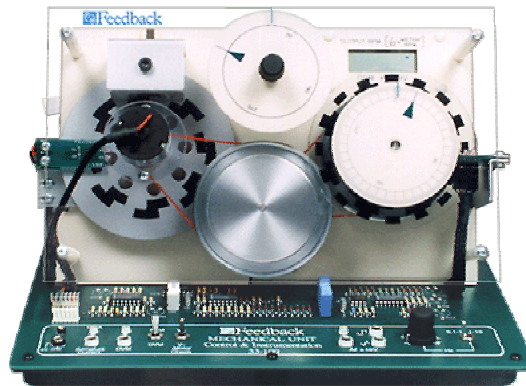


Figure 1-7: Digital Servo System

The Modular Servo system is a self contained dc servo motor system with control circuitry for the experimental study of servo motor behaviour and control. It comprises a Mechanical Unit and a Digital Unit.

The Mechanical Unit consists of a Power amplifier, dc motor, tachogenerator, absolute and incremental digital encoders, input and output analogue potentiometers, digital speed and voltage display, and a sine, square and sawtooth waveform generator.

The Digital Unit carries ADC and DAC for signal conversion, switching and multiplexing circuits, encoder output and display, linear and PWM motor drive and interface to the I/O card in the PC.



Process Control Trainer

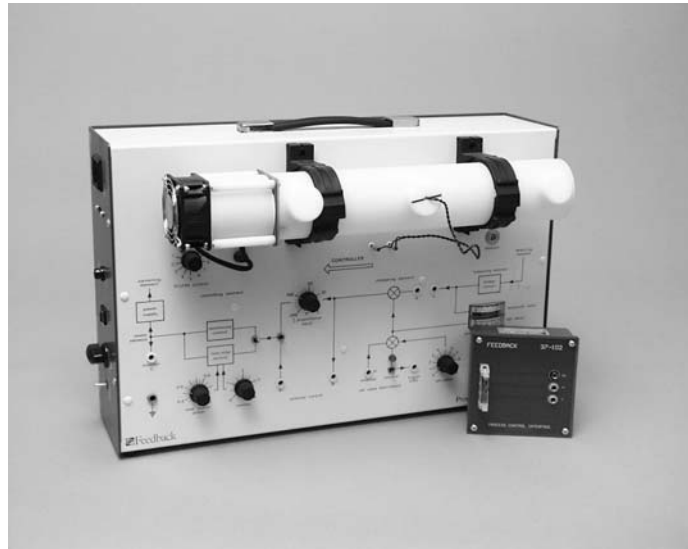


Figure 1-8: Process Control Trainer

The Process Control Trainer uses a tube equipped with a heating element at one end and a thermistor temperature detector, which can be placed in two positions in the tube. The control variable is the voltage to be applied to the heating element whilst the measured value is a voltage from the thermistor circuitry. The Process Control Trainer can be used to demonstrate and develop control techniques in a system with a transport delay.

Level and Flow process Rig

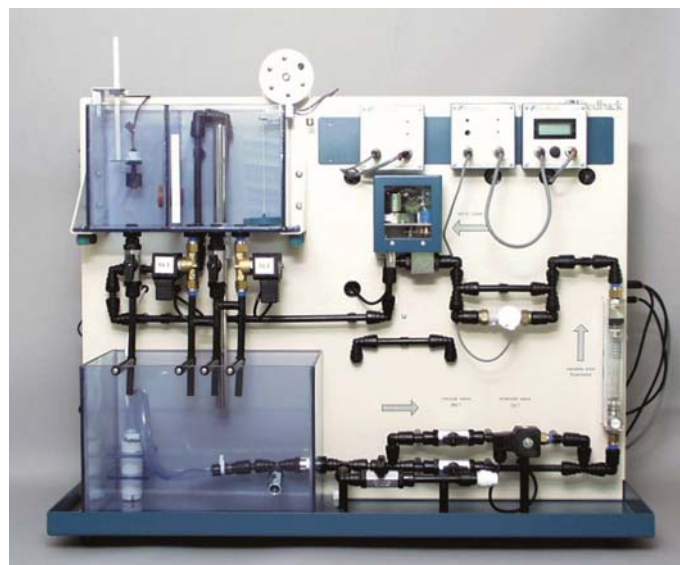


Figure 1-9: Level and Flow Process Rig



INTRODUCTION AND COMPUTER SYSTEMS INSTALLATION

Chapter 1

Introduction

The Level and Flow Process Rig uses a pump to pump water from a reservoir tank to a header tank from where it may drain back into the reservoir tank. The flow into the header tank is controlled by a servo valve operating from a 4 -20 ma range (fully closed - fully open) signal current. The header tank level transmitter consists of a float attached to a potentiometer which provides a 4-20 ma range (tank empty - tank full)input current. The 4-20 ma input and output currents are converted into 0-5 V ranges using an adjustable interface unit supplied.



INTRODUCTION AND COMPUTER SYSTEMS INSTALLATION

Chapter 1 Introduction

Notes



2 System Requirements

Hardware

The hardware required for implementing the controllers described in the accompanying manuals includes as a minimum:

Pentium 700 MHz PC with 128 MB of main memory, CD ROM drive

Either (EISA)

For all apart from Digital Servo

Advantech PCL-812PG I/O card and at least 1 free EISA slot on the motherboard

This board has the following features

- 2 digital to analogue converter channels, 0-5 V range
- 16 digital inputs
- 16 analogue to digital converters

Digital Servo only

Feedback Mica926 board and at least 1 free EISA slot on the motherboard

Or (PCI)

For all apart from Digital Servo

Advantech PCI1711 I/O card and at least one free PCI slot on the motherboard.

This board has the following features

- Windows Plug and Play device
- 2 digital to analogue converter channels, 0-5 V range
- 16 digital inputs (multiplexed to receive data from the control box)
- 16 analogue to digital converters (two used for tachometer inputs)



With the PCI1711 board a SCSI Adapter box is provided by Feedback to interface by ribbon cable to the Feedback plants.

Digital Servo only

Advantech PCI1751 I/O card and at least one free PCI slot on the motherboard.

This board has the following features

- Windows Plug and Play device
- 48 digital inputs / outputs

One or more of the Feedback Electromechanical plants previously described, with either external or built-in input/output electronics control box.

Software

Microsoft Windows 95, 98, NT, 2000

Microsoft Visual C++ Professional Edition Version 6.0

MATLAB Version 6.1 Release 12.1 from Mathworks Inc.

Simulink Version 4.1 from Mathworks Inc

Real Time Workshop Version 4.1 from Mathworks Inc

Real Time Windows Target Version 2.1 from Mathworks Inc

Feedback EISA Software Pack for MATLAB (Advantech PCL812PG or Mica926)

33-922	Digital Servo Workshop	Mica926
33-926	Precision Modular Servo	PCL812PG
33-935	Digital Pendulum	PCL812PG
33-941	Magnetic Levitation System	PCL812PG
33-948	Twin Rotor MIMO system	PCL812PG
37-903	Process Control Trainer	PCL812PG
38-905	Level and Flow	PCL812PG



Feedback PCI Software Pack for MATLAB (Advantech PCI1711 or PCI1751)

33-921	Digital Servo Workshop	PCI1751
33-927	Precision Modular Servo	PCI1711
33-936	Digital Pendulum	PCI1711
33-942	Magnetic Levitation System	PCI1711
33-949	Twin Rotor MIMO system	PCI1711
37-902	Process Control Trainer	PCI1711
38-906	Level and Flow Process Rig	PCI1711



INTRODUCTION AND COMPUTER SYSTEMS INSTALLATION

Chapter 2 System Requirements

Notes



3 Advantech PCI1711 Card

Introduction

If you have the Advantech PCI 1711 card, then follow the instructions given in this section. The PCI1711 uses the PCI bus and is therefore a Plug and Play device, which will be detected by the Windows Operating System when the computer is powered up.

Check

Check that you have the following items in the PCL-1711 Package

PCI-1711 card

CD-Rom disk

User manual

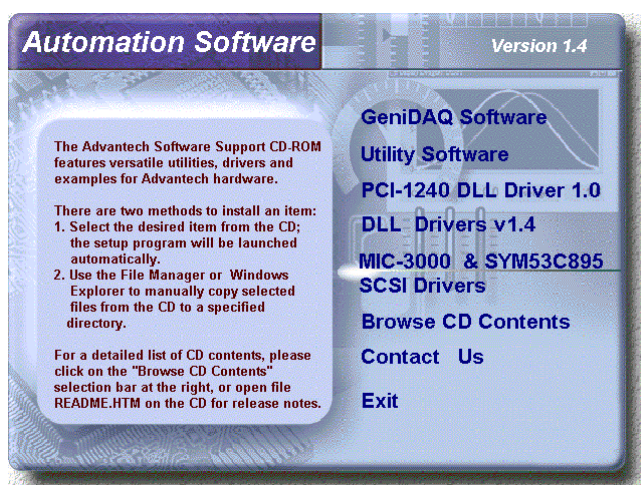
Quick start (single sheet instructions)

A) Driver Installation

Step 1

Insert the companion CD disk into your CD-ROM drive

The Setup program will be launched automatically and you will see the following Setup screen





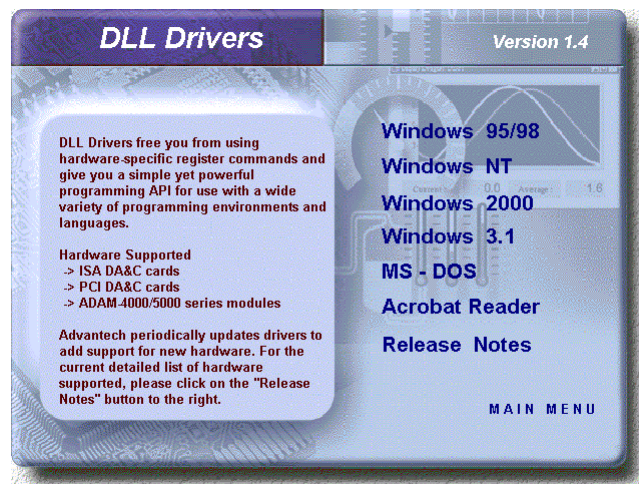
INTRODUCTION AND COMPUTER SYSTEMS INSTALLATION

Chapter 3

Advantech PCI1711 Card

Step 2

Select the **DLL Drivers** installation option. (If autoplay is not enabled, use Windows Explorer or Windows Run command to execute `setup.exe` on CD-ROM).

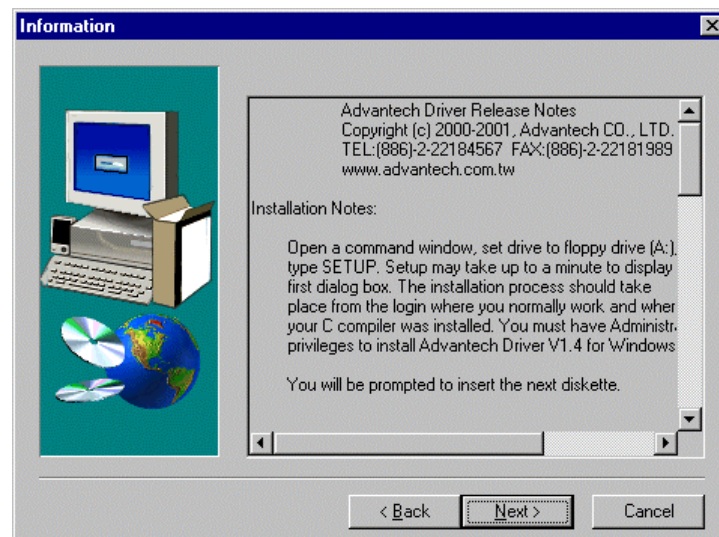


Step 3

Select your version of the **Windows** Operating System

Step 4

Follow the installation instructions step by step to complete your DLL driver setup, as shown in the screen images below

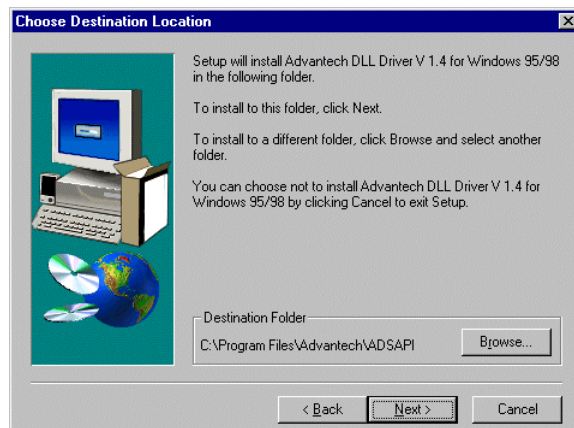
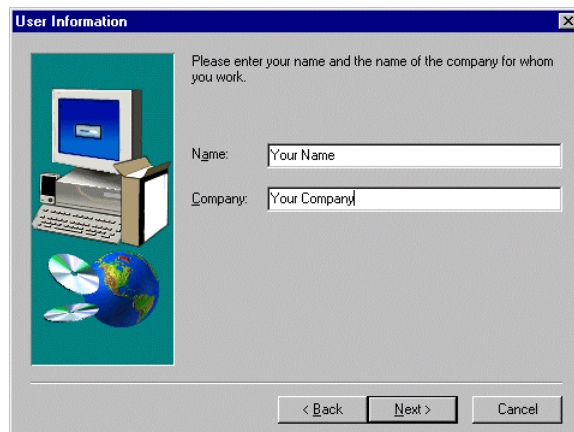
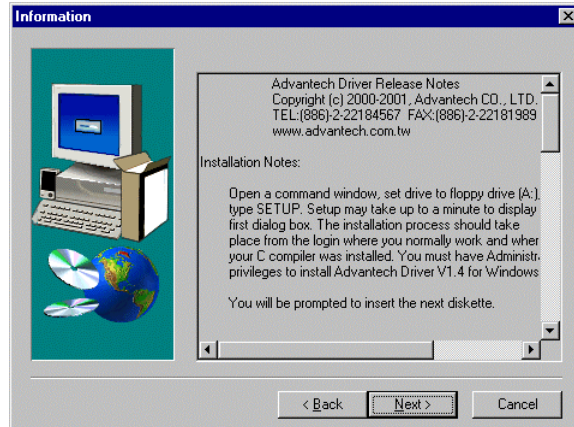




INTRODUCTION AND COMPUTER SYSTEMS INSTALLATION

Chapter 3

Advantech PCI1711 Card

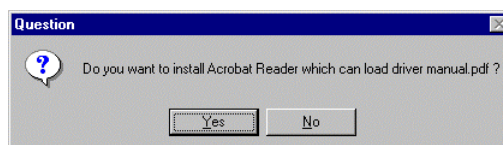
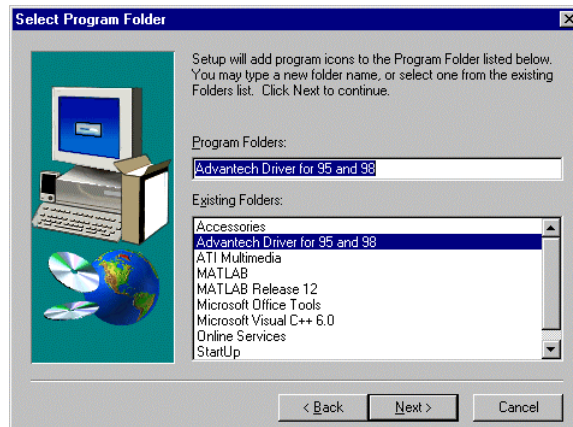
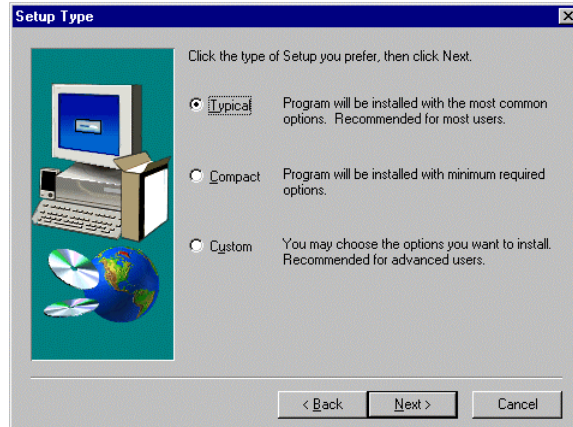




INTRODUCTION AND COMPUTER SYSTEMS INSTALLATION

Chapter 3

Advantech PCI1711 Card





B) Hardware Installation

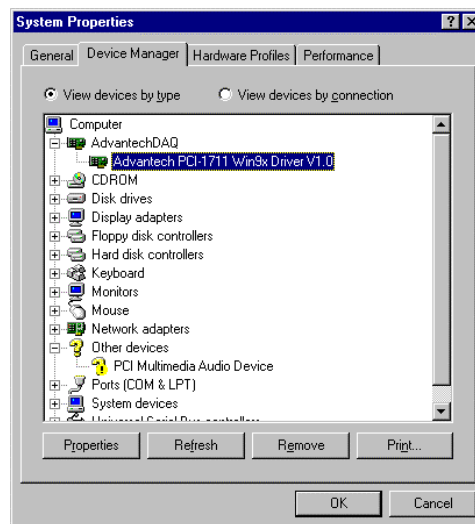
- | | |
|---------------|---|
| Step 1 | Turn off the computer and unplug the power cord and cables |
| Step 2 | Remove the cover of the computer |
| Step 3 | Remove the slot cover on the back panel of the computer |
| Step 4 | Touch the metal part of your computer to discharge static electricity on your body |
| Step 5 | Insert the PCI 1711 card into a PCI slot. Hold the card only by its edges and carefully align it with the slot, then insert the card firmly into place. Use of excessive force must be avoided otherwise the card may be damaged. |
| Step 6 | Fasten the bracket of the PCI card on the back panel rail of the computer with screws |
| Step 7 | Replace the computer cover, and re-connect the cables you removed in Step 2 |
| Step 8 | Connect the 68 pin SCSI connector on the PCI1711 board to the 68 pin connector on the Feedback SCSI Adapter box using the SCSI cable supplied. The other connectors on this box then connect to the Feedback equipment via the ribbon cables supplied |



C) Verification

Step 1 Restart your Computer. Windows will detect the presence of the PCI1711 card and should find the driver, installed in steps A)

Step 2 Access the Device Manager through the Control Panel/System/Device Manager. On the Device Manager tab of the System Property sheet you can see the Device Name of the PCI1711 listed on it.





INTRODUCTION AND COMPUTER SYSTEMS INSTALLATION

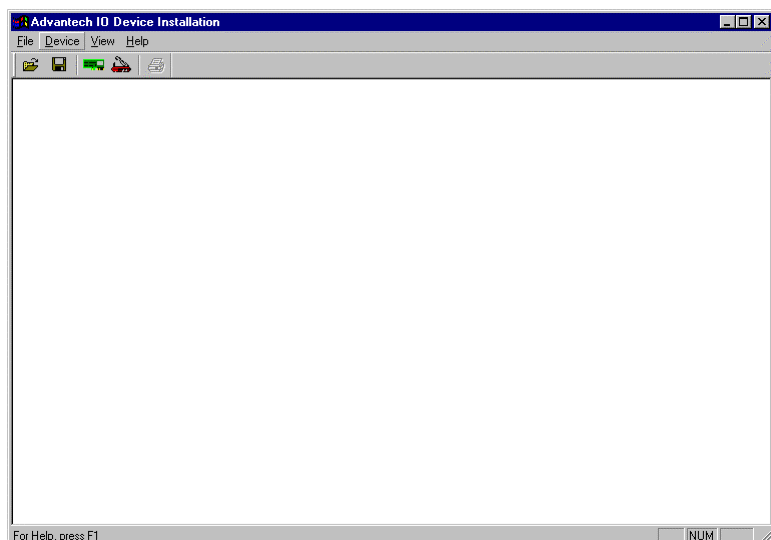
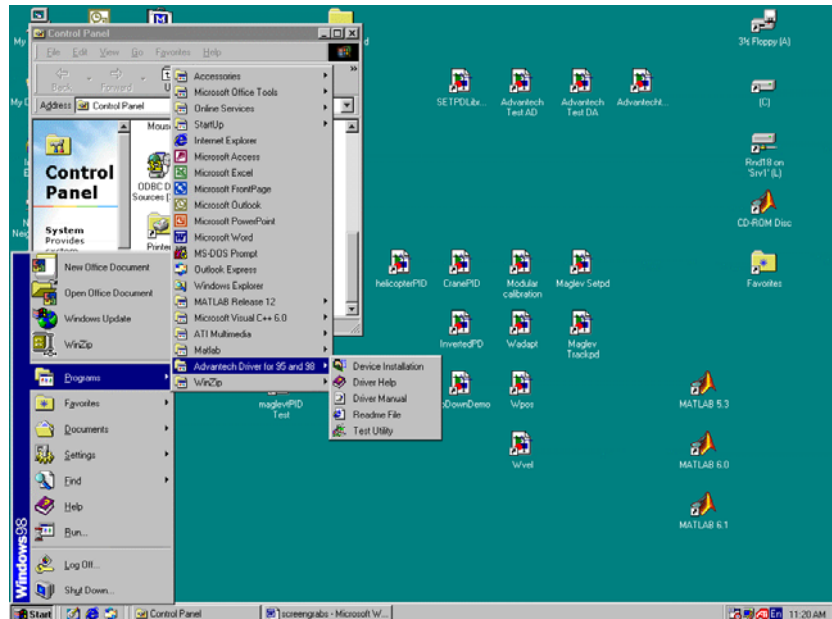
Chapter 3

Advantech PCI1711 Card

D) Device Installation

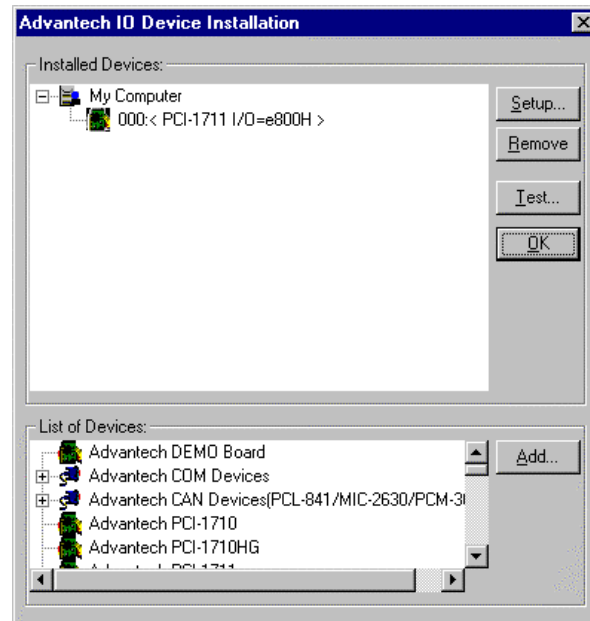
Step 1

Run the Device Installation program by accessing
Start/Programs/Advantech Driver for 95 and 98



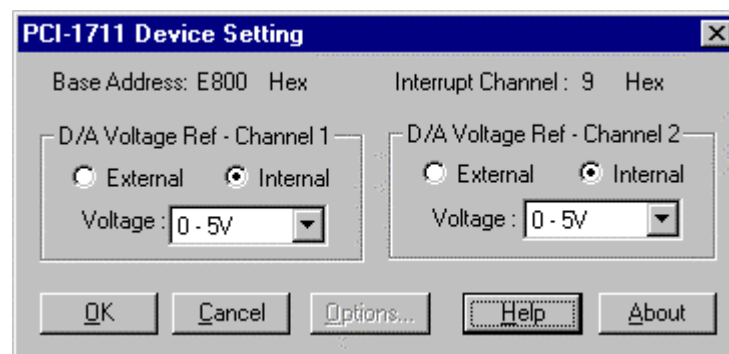
Step 2

On the Device Installation program Window, select the
Setup menu item on the menu bar, and click the **Device**
command to bring up the I/O Device Installation
dialogue box as follows



Step 3

Click the **Setup** tab. The **Device Setting** dialogue box will pop up



Step 4

Make sure the "Internal" radio button is set and that the D/A voltages are in the range 0-5 V.

Step 5

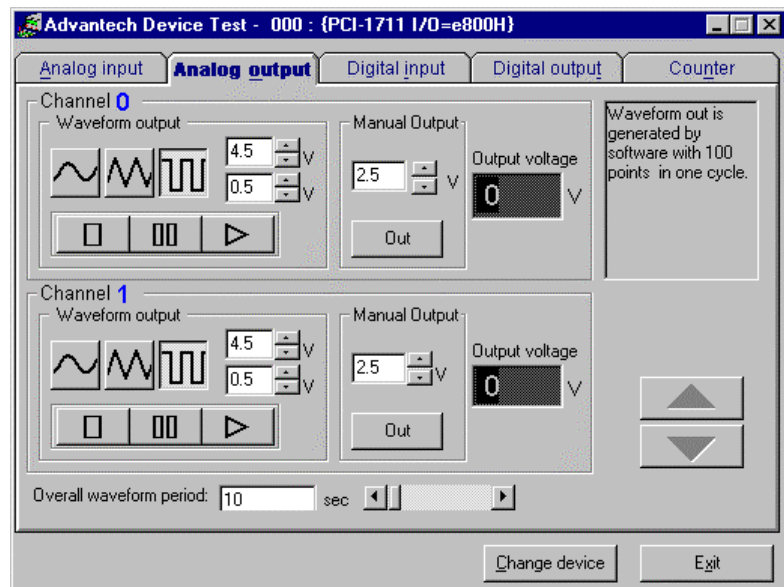
Click the **Test** button on the **Devices Installation** window. The test utility window pops up which can be used to test the analogue and digital input and output.



INTRODUCTION AND COMPUTER SYSTEMS INSTALLATION

Chapter 3

Advantech PCI1711 Card



Note

It is not necessary to make any on board adjustments to the PCI1711 board as its base address has automatically been configured by Windows during the Plug and play setup.



**INTRODUCTION AND
COMPUTER SYSTEMS INSTALLATION**

Chapter 3

Advantech PCI1711 Card

Notes



4 Advantech PCL- 812PG card (EISA)



The PCL-812PG contains components which can be damaged by static electricity. Before removing the PCL-812 board from its conductive packaging, touch your hand on a grounded metal object such as the case of your PC. Avoid touching the components on the board or the plated edge connectors.

Before installing the PCL-812PG interface card into your computer make sure that the switches and jumpers shown in Figure 4-1 are set correctly, as detailed in this section.

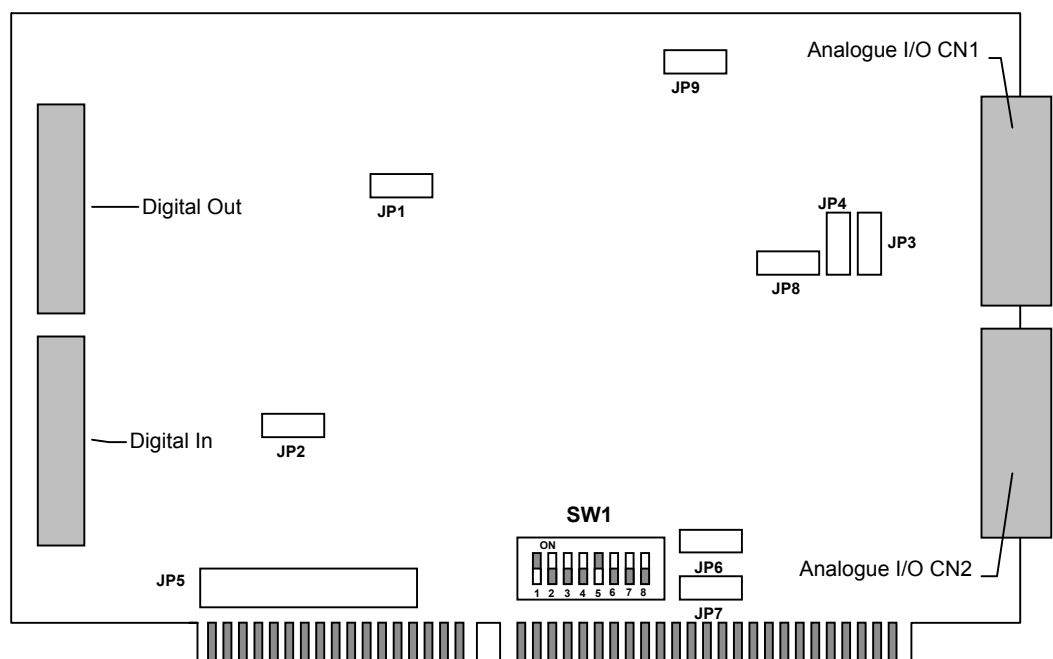


Figure 4-1: 16 bit PCL-812PG Interface Card - Switch and Jumper positions

Note

The board shown above is a 16 bit short board. Some users may have the 8 bit full length PCL- 812 board for which the SW1 switch settings are different. If you have this board refer to the settings in the following section for the 8 bit board. Other settings and cable connections are the same for both boards.



Switch settings - 16 bit board

Switch SW1 controls the base address for I/O transfer (switches 1-6), and the CPU wait states (Switches 7-8). They should be set as follows.

For convenience, both decimal and hex addresses are shown in the table below

SW1 Base Address Selection

I/O Address Range (hex and decimal)	SW1 Switch position					
	1	2	3	4	5	6
200-20F (512-527 Dec)	OFF	ON	ON	ON	ON	ON
210-21F (528-543 Dec)	OFF	ON	ON	ON	ON	OFF
220-22F (544-559 Dec) *	OFF	ON	ON	ON	OFF	ON
230-23F (560-575 Dec)	OFF	ON	ON	ON	OFF	OFF
300-30F (768-783 Dec)	OFF	OFF	ON	ON	ON	ON
3F0-3FF (1008-1023 Dec)	OFF	OFF	OFF	OFF	OFF	OFF

* Factory default setting



Note that some video cards or floppy disk drives use the higher base addresses such as 1008 (Hex 3F0), and should be avoided if possible.

SW1 Wait State Selection

Time Delay	SW1 Switch Settings	
	7	8
0 *	ON	ON
2	OFF	ON
4	ON	OFF
6	OFF	OFF

* Factory default setting



Jumper Settings

The Jumpers on the PCL-812 board should be set in the positions indicated below.

JP1 - Trigger Source Selection

	JP1
INT	☐
TRG	☐
EXT	☐

JP2 - Counter Input Clock Selection

	JP2
INT	☐
CLK	☐
EXT	☐

JP3, JP4 - D/A Reference Selection

JP3	☐	☐	☐	D/A1
JP4	☐	☐	☐	D/A2
	INT	VREF	EXT	

JP5 - IRQ Level Selection

2	3	4	5	6	7	X
☐	☐	☐	☐	☐	☐	☐
☐	☐	☐	☐	☐	☐	☐
JP5						

JP6, JP7 - DMA Channel Selection

1	3	X	1	3	X
☐	☐	☐	☐	☐	☐
☐	☐	☐	☐	☐	☐
JP6			JP7		



JP8, JP9 - D/A Internal Reference, A/D Maximum Input Voltage Selection

	10	5	
JP8	O	O	O
JP9	O	O	O

Depending on the specification and configuration of your PC you may need to change the Base Address and Wait State settings.



None of the other settings need to be changed.

Switch SW1 settings for 8 bit full length PCL - 812 board

I/O Address Range (hex and decimal)	SW1 Switch position					
	1	2	3	4	5	6
200-20F (512-527 Dec)	ON	ON	ON	ON	ON	X
210-21F (528-543 Dec)	ON	ON	ON	ON	OFF	X
220-22F (544-559 Dec) *	ON	ON	ON	OFF	ON	X
230-23F (560-575 Dec)	ON	ON	ON	OFF	OFF	X
300-30F (768-783 Dec)	OFF	ON	ON	ON	ON	X
3F0-3FF (1008-1023 Dec)	OFF	OFF	OFF	OFF	OFF	X

Note: An "X" in the table above means that this switch is not used.



Installation of the PCL-812PG card and cables

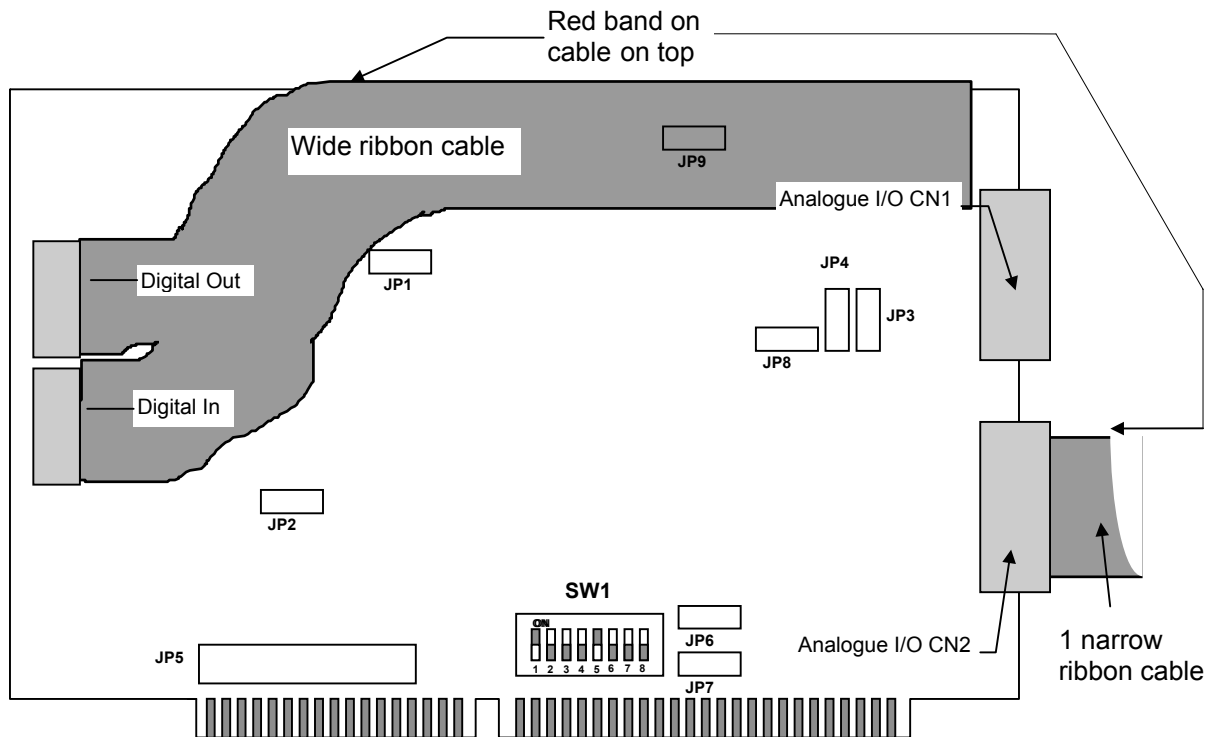
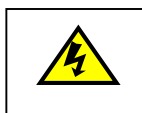


Figure 4-2: PCL- 812 Interface Card Ribbon Cable connections

Two slots are required at the rear of the PC - one for the two 20-way analogue port sockets, and one for the 40-way socket at the end of the wide ribbon cable for the digital ports. Use screws to secure the card and ribbon socket plates to the casing of the PC.

Carefully install the interface card in a spare slot on the PC motherboard (or in a slot on the passive backplane if your PC has this type of architecture - common with some 19 inch rack mounted PC's). If in any doubt as to the correct procedure consult the documentation supplied with the PC.



**Turn OFF and disconnect the mains power supply
BEFORE opening the computer**

Before opening the computer it is important to ensure that the computer is switched off and that the mains power supply is not connected.



INTRODUCTION AND COMPUTER SYSTEMS INSTALLATION

Chapter 4 Advantech PCL 812PG Card (EISA)

Once you have opened the computer, check to see if there are any other cards installed, for example a network or sound card. It is important to determine the addresses of any such cards as your Feedback System will not function correctly if there is another card using all, or part of, the address space assigned to the PCL-812PG interface card.



5 Feedback Mic926 (For Digital Servo 33-004 only)



The Feedback Mic926 contains components which can be damaged by static electricity. Before removing the Mica 926 board from its conductive packaging, touch your hand on a grounded metal object such as the case of your PC. Avoid touching the components on the board or the plated edge connectors.

This part of the system resides inside the computer and is plugged into a spare slot in the computer expansion bus.

The computer interface (type Mic926) should be installed as follows:

Check the user options on the interface. These are set by dual in line switches and some jumpers.

The correct settings are:

Interrupt processing	OFF
Mode (16-way link)	IBM
Base Address (4 di1 switches)	All Open

Table 5-1: Default MICA settings

The interface should then be installed in a spare slot in the computer expansion area. The Mic926 only needs an 8 bit (short) slot, but will work in a 16 bit (long) slot if necessary. The expansion area is accessed by removing the cover from the computer. Turn off all power to the computer and any peripheral devices before you attempt to install the interface. If a spare slot is not available then an existing card must be removed. If you remove the video adapter, I/O port or disk controller your system will not work.

A small amount of resistance will be felt when pushing the interface into the expansion slot, do not force it in or damage may occur. When it is located correctly secure it with a screw in the socket bracket like the other cards in the expansion area.

If the computer does not function properly, there is no



communication between the computer and the hardware. First check all power supplies.

If this is not the problem then there may be a conflict between the Mic926 address and an existing card in the expansion area. For example some network cards use the same address as the default setting on the Mic926.

There are two alternative solutions to this problem. One is to change the address of the conflicting card. The card would be identified by removing the other interface cards in the expansion area and trying to run the system. Once identified then most card addresses can be changed with hardware or software switches.

The other solution is to change the address of the Mic926. This is done by removing the card and changing the setting of the 4-way dual in line switch.

The available addresses are shown in

Switch Weight	4	16	8	not used
Base Address	sw 1.1	sw 1.2	sw 1.3	sw 1.4
768	open	open	open	
772	closed	open	open	
776	open	open	closed	
780	closed	open	closed	
784	open	closed	open	
788	closed	closed	open	
792	open	closed	closed	
ILLEGAL	closed	closed	closed	

Table 5-2: Base Address Switch Settings



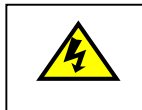
INTRODUCTION AND COMPUTER SYSTEMS INSTALLATION

Chapter 5

Feedback Mic926

The ability to change the Mic926 address means that it is possible to have two Mic926 interfaces in a computer at one time. This could be used to have two products connected and both sets of software loaded. One interface would have to be on a different address to the other.

Carefully install the interface card in a spare slot on the PC motherboard (or in a slot on the passive backplane if your PC has this type of architecture - common with some 19 inch rack mounted PC's). If in any doubt as to the correct procedure consult the documentation supplied with the PC.



**Turn OFF and disconnect the mains power supply
BEFORE opening the computer**

Before opening the computer it is important to ensure that the computer is switched off and that the mains power supply is not connected.

Once you have opened the computer, check to see if there are any other cards installed, for example a network or sound card. It is important to determine the addresses of any such cards as your Feedback System will not function correctly if there is another card using all, or part of, the address space assigned to the Mic926 interface card.



**INTRODUCTION AND
COMPUTER SYSTEMS INSTALLATION**

Chapter 5
Feedback Mic926

Notes



6 Advantech PCI 1751 Card (for Digital Servo 33-004 only)

Introduction

If you have the Advantech PCI 1751 card, then follow the instructions given in this section. The PCI1751 uses the PCI bus and is therefore a Plug and Play device, which will be detected by the Windows Operating System when the computer is powered up.

Check

Check that you have the following items in the PCL-1751 Package

PCI-1751 card

CD-Rom disk

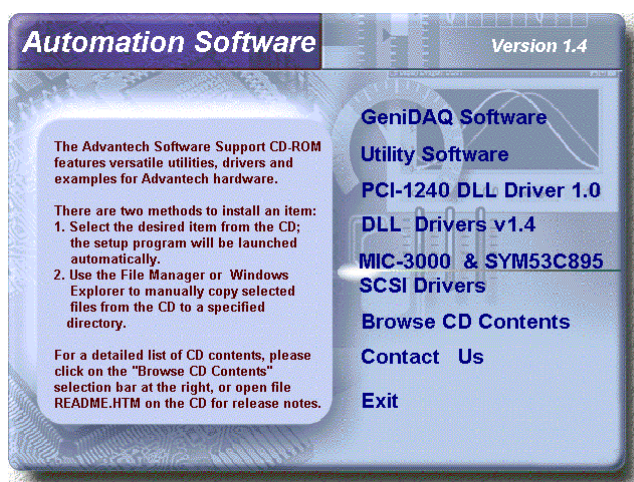
User manual

A) Driver Installation

Step 1

Insert the companion CD disk into your CD-ROM drive

The Setup program will be launched automatically and you will see the following Setup screen



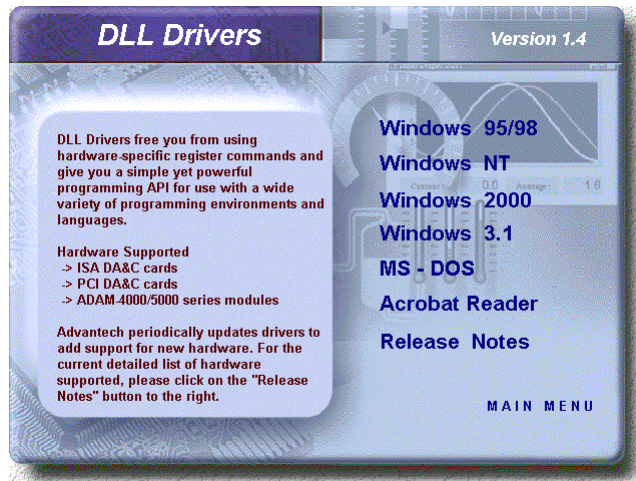


INTRODUCTION AND COMPUTER SYSTEMS INSTALLATION

Chapter 6 Advantech PCI1751 Card (for Digital Servo 33-004 only)

Step 2

Select the **DLL Drivers** installation option. (If autoplay is not enabled, use Windows Explorer or Windows Run command to execute `setup.exe` on CD-ROM).

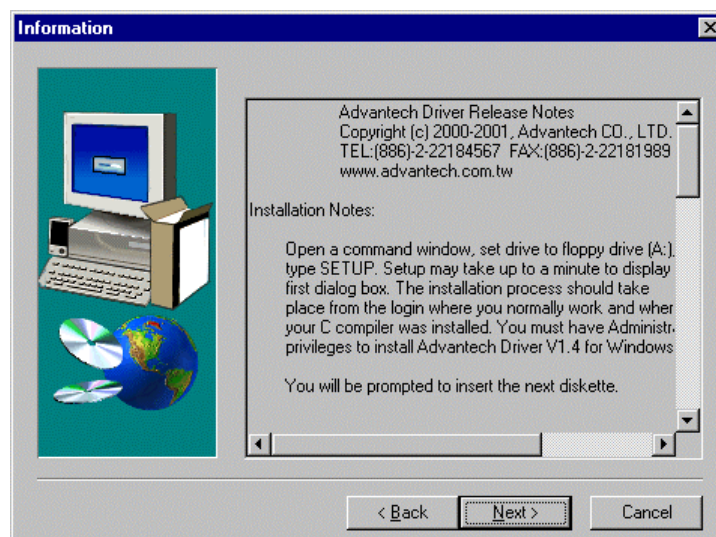


Step 3

Select the **Windows 95/98** or the option according to your operating system. **Windows NT**

Step 4

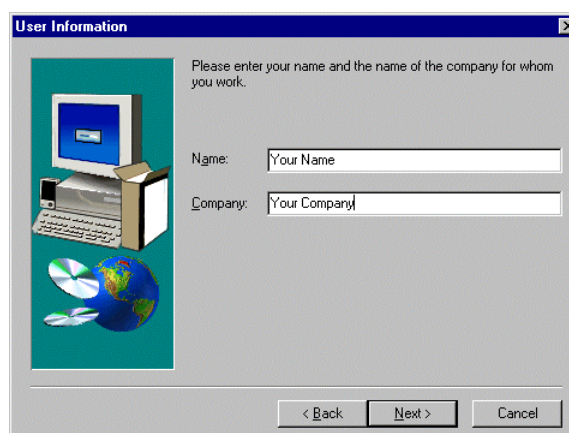
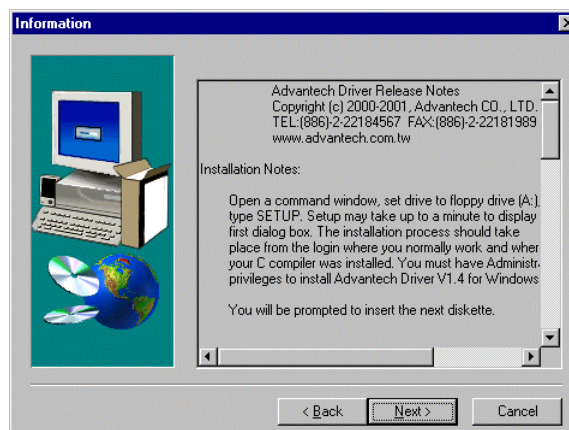
Follow the installation instructions step by step to complete your DLL driver setup, as shown in the screen images below





INTRODUCTION AND COMPUTER SYSTEMS INSTALLATION

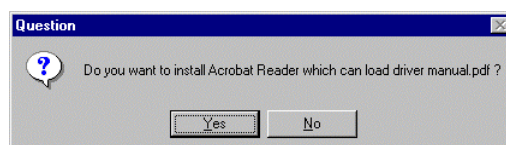
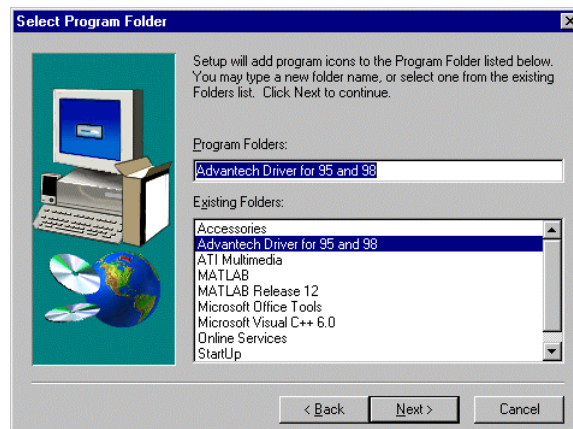
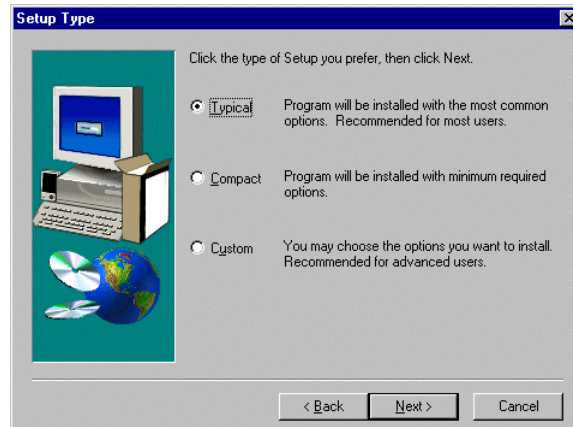
Chapter 6 Advantech PCI1751 Card (for Digital Servo 33-004 only)





INTRODUCTION AND COMPUTER SYSTEMS INSTALLATION

Chapter 6 Advantech PCI1751 Card (for Digital Servo 33-004 only)





B) Hardware Installation

- | | |
|---------------|---|
| Step 1 | Turn off the computer and unplug the power cord and cables |
| Step 2 | Remove the cover of the computer |
| Step 3 | Remove the slot cover on the back panel of the computer |
| Step 4 | Touch the metal part of your computer to discharge static electricity on your body |
| Step 5 | Insert the PCI 1751 card into a PCI slot. Hold the card only by its edges and carefully align it with the slot, then insert the card firmly into place. Use of excessive force must be avoided otherwise the card may be damaged. |
| Step 6 | Fasten the bracket of the PCI card on the back panel rail of the computer with screws |
| Step 7 | Replace the computer cover, and re-connect the cables you removed in Step 2 |
| Step 8 | Connect the 68 pin SCSI connector on the PCI1751 board to the connector on the Digital Servo Digital Board using the special cable supplied by Feedback. |



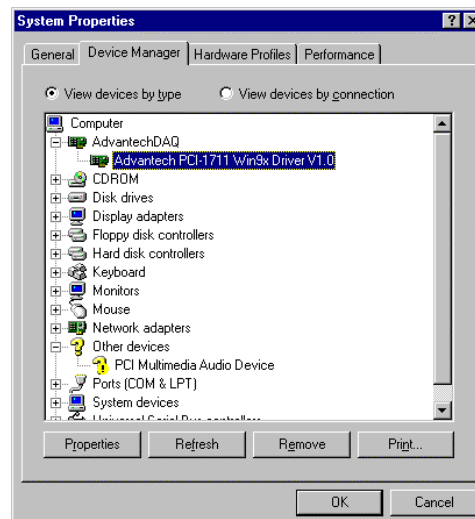
INTRODUCTION AND COMPUTER SYSTEMS INSTALLATION

Chapter 6 Advantech PCI1751 Card (for Digital Servo 33-004 only)

C) Verification

Step 1 Restart your Computer. Windows will detect the presence of the PCI1751 card and should find the driver, installed in steps **A)**

Step 2 Access the Device Manager through the Control Panel/System/Device Manager. On the Device Manager **tab** of the System Property sheet you can see the Device Name of the PCI1751 listed on it.





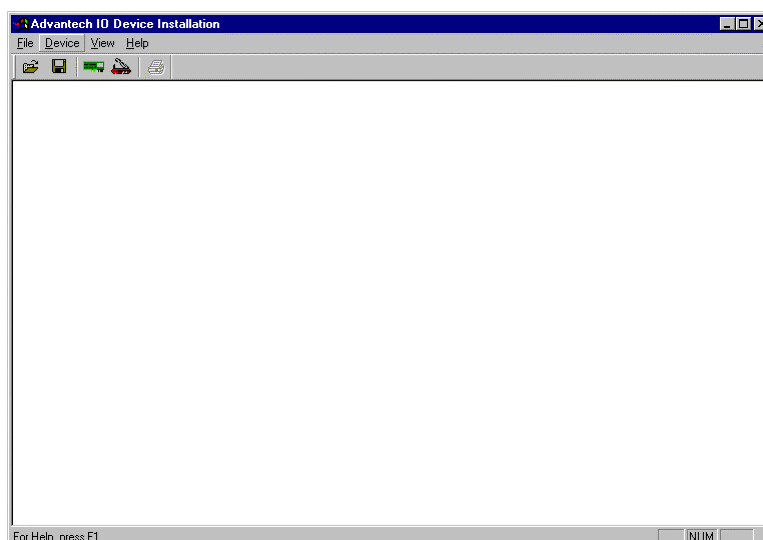
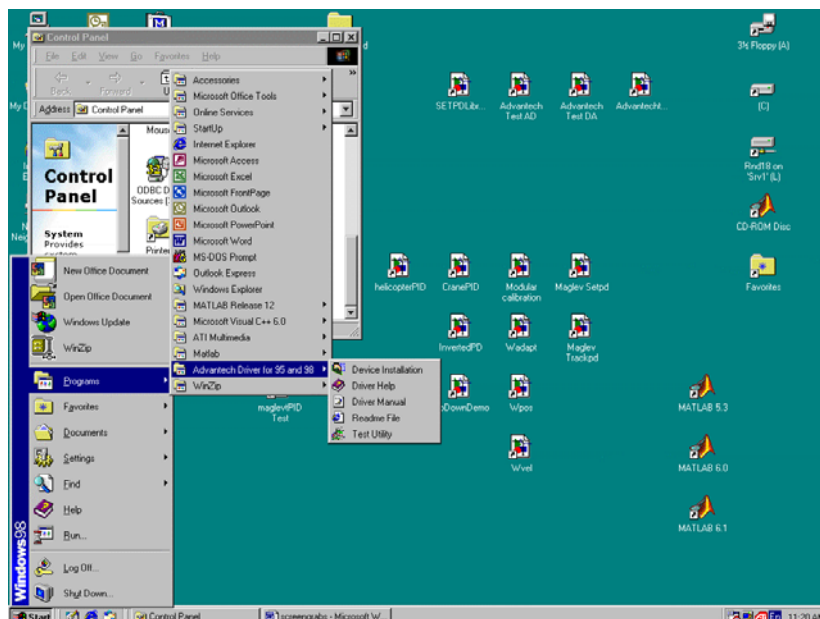
INTRODUCTION AND COMPUTER SYSTEMS INSTALLATION

Chapter 6 Advantech PCI1751 Card (for Digital Servo 33-004 only)

D) Device Installation

Step 1

Run the Device Installation program by accessing
Start/Programs/Advantech Driver for 95 and 98



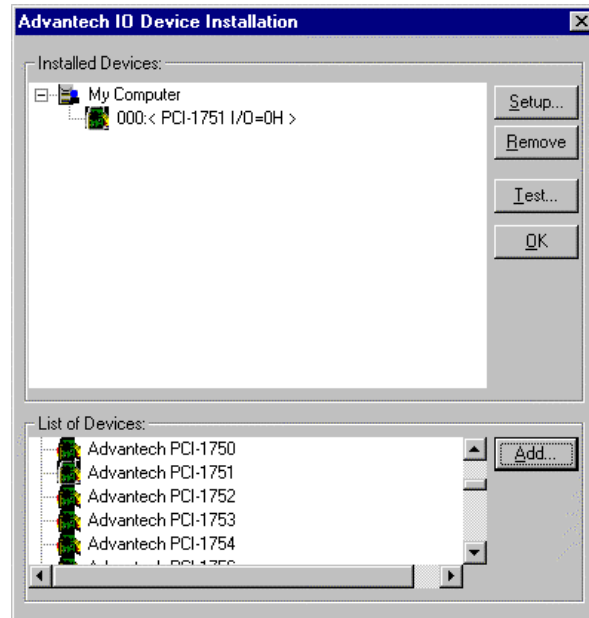
Step 2

On the Device Installation program Window, select the Setup menu item on the menu bar, and click the **Device** command to bring up the I/O Device Installation dialogue box as follows



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Chapter 6 Advantech PCI1751 Card (for Digital Servo 33-004 only)



The window shown above shows that the board has been correctly installed.

Note

It is not necessary to make any on board adjustments to the PCI1751 board as its base address has automatically been configured by Windows during the Plug and play setup.



7 Software

Installation

Follow this sequence of instructions to install all the systems

Hardware

- 1 Assemble, commission and test the Feedback Electromechanical plant according to the instructions contained in the following Feedback Installation and Commissioning manuals. An Advantech I/O board should already have been installed as described in Chapter 3 (Advantech PCI1711) or Chapter 4 (Advantech PCL-812PG)

33-921/922-1	for the Digital Servo
33-935/936-0	for the Digital Pendulum
33-941/942-0	for Magnetic Levitation
33-948/949-0	for the Helicopter TRMS
33-926/927-0	for the Modular Servo
37-902/903-1	for the Process Trainer
38-905/906-1	for the Level and Flow Rig

MATLAB Software

- 2 Install MATLAB, Simulink, Real Time Workshop, Real Time Windows Target, and Microsoft Visual C++ according to the supplier's instructions.
- 3 Start MATLAB
- 4 To set up Real Time Windows Target; in the MATLAB Command Window type:

```
rtwintgt -setup
```

Note space before the "-"
- 5 To inform MATLAB of the location of your Visual C++ compiler; in the MATLAB Command window type:

```
mex -setup
```

Note space before the "-"

MATLAB responds with

```
Would you like mex to locate installed  
compilers [y]/n?
```

In the MATLAB Command window type "y"



MATLAB responds similarly to the following:

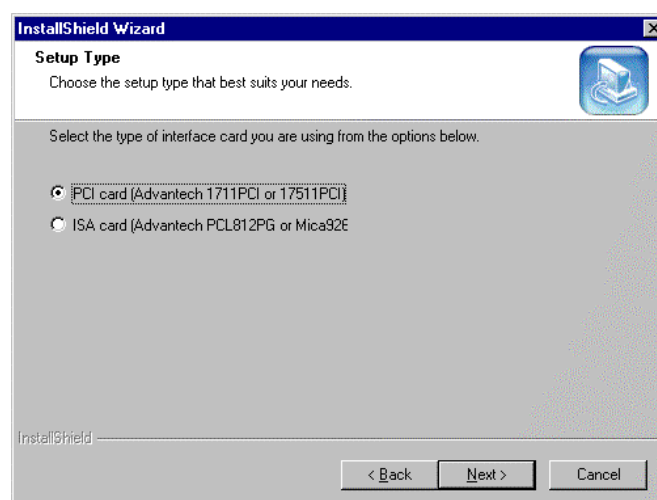
```
Please choose your compiler for building
external interface (MEX) files:
Select a compiler:
[1] Digital Visual Fortran version 6.0 in
C:\Program Files\Microsoft Visual Studio
[2] Lcc C version 2.4 in
C:\MATLABR12\sys\lcc
[3] Microsoft Visual C/C++ version 6.0 in
C:\Program Files\Microsoft Visual Studio
```

In the MATLAB Command window type "3" to select Visual C++

**Feedback
Software
for
MATLAB**

6. Insert the Feedback software CD for the particular Feedback plant you are installing.

After some preliminary screens you will be asked to choose the type of I/O board. Select the correct one by clicking the radio button as shown in the figure below. The one shown is for the Digital Servo - other systems will have similar screens.



Click Next

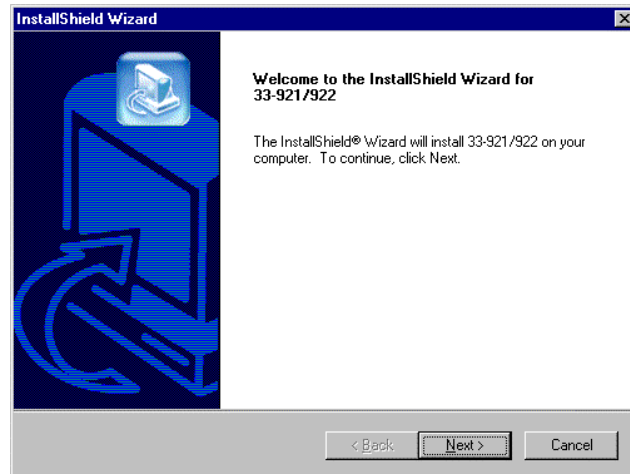
The following screen is shown



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Chapter 7

Software



Click **Next**

Your system is now installed

Note

If you have previously installed the model files and have made any changes to the Simulink models in the C:\MatlabR12\Feeback folder, you should first save the contents of C:\MatlabR12\Feeback to another folder, otherwise they will be overwritten.



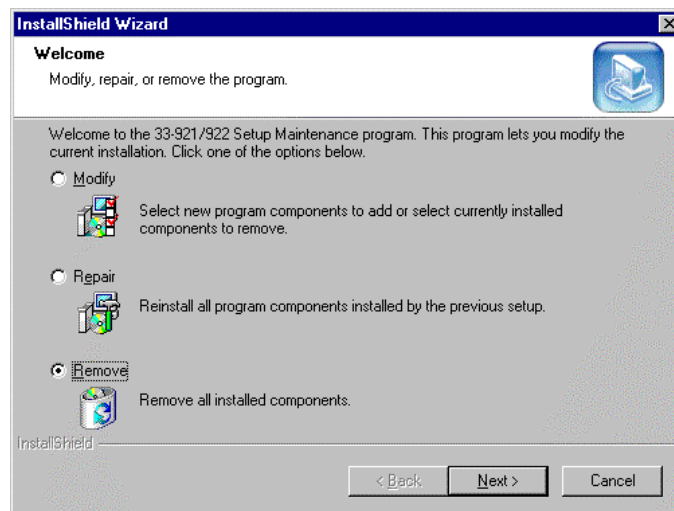
Un-installation

Feedback
software for
MATLAB
un-installation

To uninstall all the Feedback base software files, follow the following sequence:

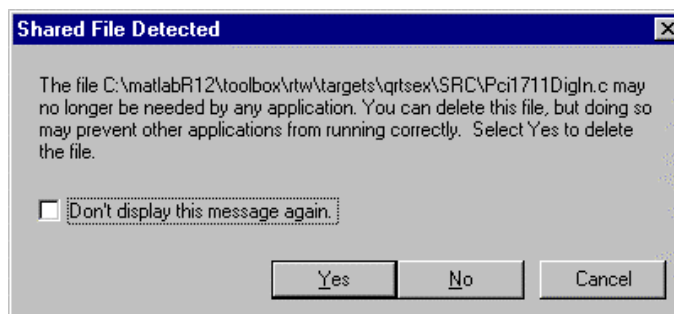
Insert the Feedback software CD for the particular Feedback plant you are un-installing.

You will see the following screen



Click on the Remove all installed components radio button.

The following screen will be shown



You will be asked to confirm the deletion as shown in the screen above.

Click on the Yes button if you wish to delete this file.



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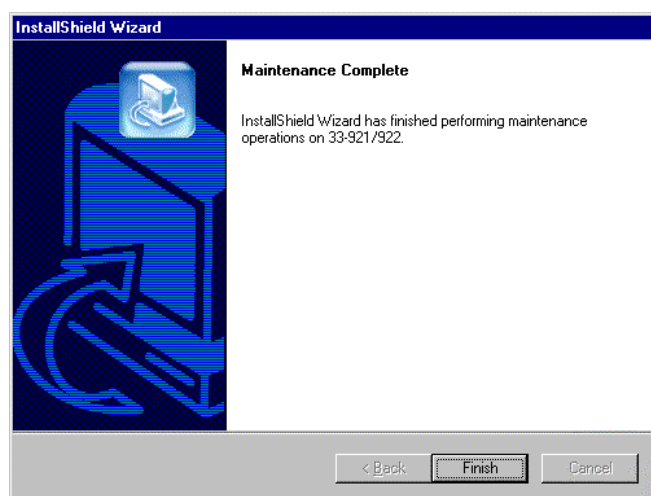
Chapter 7

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You will get a similar request for all other files which may be shared with other applications. Respond with **Yes** if you wish to delete them.

Note the contents of the above messages, these files may be shared and deletion may stop other programs from working

The following screen appears



Click on **Finish**

The Feedback software for MATLAB will be removed for this particular system.

Note

There are two other radio buttons shown on the installation screen marked **Modify** and **Repair**. It is recommended that you do not use these. Instead to re-install a system, first **Un-install** and then **Install**. Remember to first save any of your modified Simulink models.



Files placed on disk after software installation

Base software

After the software installation has been completed as described above, the following files should now be present on your computer.

Directory Name	Contents
matlabroot\toolbox\rtw\targets	Folder fbk
matlabroot\toolbox\rtw\targets\fbk	Folders fbk, src
matlabroot\toolbox\rtw\targets\fbk\src	All I/O drivers
matlabroot\toolbox\rtw\targets\fbk\fbk	Feedback I/O additions to Simulink Library
matlabroot\work	start.m - M file containing the Path statements for MATLAB including the Feedback special files

Table 7-1: Location of Feedback base software

The following paths will be added to the MATLAB Path Browser window.

```
matlabroot\toolbox\rtw\targets\fbk
matlabroot\toolbox\rtw\targets\fbk\lib
matlabroot\toolbox\rtw\targets\fbk\src
matlabroot\toolbox\rtw\targets\fbk\fbk
```

Applications software

After the applications software installation has been completed as described above, the following Simulink model files should now be present on your computer.

Directory Name	Contents
matlabroot\feedback\pendulum	CranePID.mdl InvertedPD.mdl UpDownDemo.mdl
matlabroot\feedback\helicopter	helicopterPID.mdl
matlabroot\feedback\maglev	calibration.mdl, Setpd.mdl Trackpd.mdl
matlabroot\feedback\modular	calibration.mdl, Wadapt.mdl Wpos.mdl Wvel.mdl
matlabroot\feedback\servotrainer	EISA - MicaServoIODemo.mdl MicaStMotor.mdl, ServoMicaPID.mdl PCI - ServoIODemo.mdl, StMotor.mdl, ServoPID.mdl
matlabroot\feedback\levelflow	levelflowcalib.mdl, levelflowonoff.mdl, levelflowpid



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Directory Name	Contents
matlabroot\feedback\ptrainer	ptcalibration.mdl, ptraineronoff.mdl, ptrainerpid.mdl

Table 7-2: Location of Feedback Simulink Application models

Note

matlabroot is the general name of the directory, where you have chosen to locate your copy of Matlab

Note also that the Simulink applications modules in Table 7-2 are either EISA or PCI versions, but **not** both on the same PC



**INTRODUCTION AND
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